

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Practice Program

Date: 27/08/2026

Week No.: 7

Program S. No. 13

Title	Interactive Scatter Plot with Bokeh			
Problem Statement	Create an interactive scatter plot showing the relationship between petal length and petal width from the Iris dataset using Bokeh.			
Objectives	• To understand interactive scatter plot visualization using Bokeh. • To load and organize the Iris dataset using Pandas and scikit-learn. • To visualize the relationship between petal length and petal width. • To differentiate Iris species using separate colors and an interactive legend. • To use Bokeh hover, zoom, pan, reset and save tools for interactive analysis.			
Dataset URL Link / File	Dataset: Built-in Iris dataset loaded using sklearn.datasets.load_iris(). Selected attributes: Petal Length (cm), Petal Width (cm), and Species.			
	Species		Samples	
	setosa		50	
	versicolor		50	
	virginica		50	
Functions Details	Sl. No.	Function	Package	Description of function
	1	load_iris()	sklearn	Loads the built-in Iris dataset.
	2	rename()	Pandas	Renames dataset columns for clearer labels.
	3	map()	Pandas	Maps numeric target values to species names.
	4	DataFrame()	Pandas	Represents the selected Iris attributes in tabular form.
	5	ColumnData Source()	Bokeh	Creates a Bokeh-friendly data source.
	6	figure()	Bokeh	Creates the interactive plotting canvas.
	7	scatter()	Bokeh	Plots petal length against petal width.
	8	HoverTool()	Bokeh	Displays species and measurements on hover.
	9	add_tools()	Bokeh	Adds the hover tool to the Bokeh figure.
	10	value_counts()	Pandas	Counts the records belonging to each Iris species.
	11	corr()	Pandas	Calculates correlation between petal length and width.
	12	output_file()/save()	Bokeh	Saves the interactive visualization as an HTML file.

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Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import Pandas, the Iris dataset utility, and required Bokeh modules. 2. Load the Iris dataset and rename the petal measurement columns for clarity. 3. Convert numeric target values into the species names setosa, versicolor and virginica. 4. Create a DataFrame containing Petal Length, Petal Width and Species. 5. Display sample records, dataset shape, species counts and correlation for verification. 6. Create a Bokeh figure with interactive pan, zoom, reset and save tools. 7. Plot one scatter series for each Iris species using a ColumnDataSource. 8. Add HoverTool so that species and measurements appear when the pointer is placed over a point. 9. Enable an interactive legend so species can be hidden or displayed. 10. Save the interactive Bokeh chart as an HTML file and verify successful execution. Flow: START -> Load Iris Dataset -> Prepare DataFrame -> ColumnDataSource -> Bokeh Figure -> Scatter Plot -> Hover/Legend -> Save HTML -> STOP
Program	<pre>import pandas as pd from sklearn.datasets import load_iris from bokeh.plotting import figure, output_file, save from bokeh.models import ColumnDataSource, HoverTool from bokeh.palettes import Category10 # Load the Iris dataset iris = load_iris(as_frame=True) df = iris.frame.rename(columns={ "petal length (cm)": "Petal Length (cm)", "petal width (cm)": "Petal Width (cm)" }) df["Species"] = df["target"].map(dict(enumerate(iris.target_names))) df = df[["Petal Length (cm)", "Petal Width (cm)", "Species"]] print(df.head(10)) print("Dataset Shape:", df.shape) print("Species Counts:") print(df["Species"].value_counts()) correlation = df["Petal Length (cm)"].corr(df["Petal Width (cm)"]) print("Correlation:", round(correlation, 3)) # Create the interactive Bokeh figure p = figure(title="Interactive Scatter Plot: Iris Petal Length vs Petal Width", x_axis_label="Petal Length (cm)", y_axis_label="Petal Width (cm)",</pre>

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```
width=900,
height=520,
tools="pan,wheel_zoom,box_zoom,reset,save"
)

colors = Category10[3]
for color, species in zip(colors, iris.target_names):
    subset = df[df["Species"] == species]
    source = ColumnDataSource(subset)
    p.scatter(
        x="Petal Length (cm)",
        y="Petal Width (cm)",
        source=source,
        size=9,
        color=color,
        alpha=0.75,
        legend_label=species
    )

hover = HoverTool(tooltips=[
    ("Species", "@Species"),
    ("Petal Length", "@{Petal Length (cm)}{0.0} cm"),
    ("Petal Width", "@{Petal Width (cm)}{0.0} cm")
])
p.add_tools(hover)
p.legend.title = "Species"
p.legend.location = "top_left"
p.legend.click_policy = "hide"

output_file("program13_iris_scatter.html")
save(p)
print("Interactive scatter plot saved successfully.")
```

**Output
Snapshots
With
Explanation/
Observations**

The following outputs were generated by executing the complete Program 13 code. The static chart image represents the interactive Bokeh visualization; the HTML file retains hover, zoom, pan and reset functionality.

Jupyter Output - Iris Dataset Summary

```
First 10 rows: Petal Length (cm), Petal Width (cm), Species
1.4  0.2  setosa
1.4  0.2  setosa
1.3  0.2  setosa
1.5  0.2  setosa
1.4  0.2  setosa
...
Dataset Shape: (150, 3)
Species Counts: setosa 50 | versicolor 50 | virginica 50
Correlation (Petal Length vs Petal Width): 0.963
```

Fig. 1.1: Iris Dataset and Correlation Summary

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The output verifies that 150 Iris samples were loaded, with 50 records for each species. The petal length-petal width correlation is approximately 0.963, indicating a strong positive relationship.

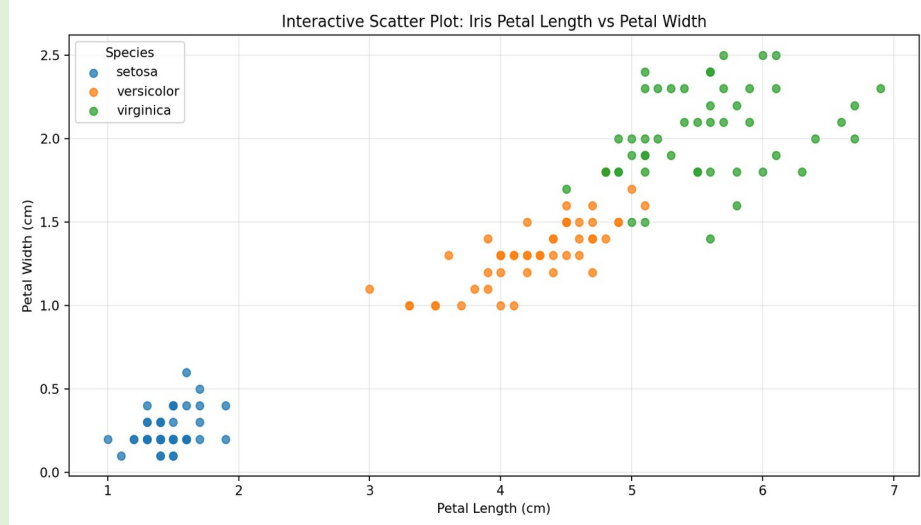


Fig. 1.2: Interactive Scatter Plot Output

The scatter plot shows three clearly distinguishable Iris species. Setosa forms a compact cluster at lower petal measurements, while versicolor and virginica occupy progressively larger petal-length and petal-width ranges.

Jupyter Output - Program 13 Final Status

```
Interactive scatter plot created successfully using Bokeh.
HTML file saved: program13_iris_scatter.html
Total records visualized: 150
Species displayed: setosa, versicolor, virginica
Hover, zoom, pan, reset and save tools are enabled.
```

Fig. 1.3: Program 13 Execution and HTML Save Confirmation

The final execution confirms successful generation of the interactive Bokeh scatter plot and its HTML output. Hover, zoom, pan, reset and save tools are available in the saved interactive file.

Conclusions

The experiment was completed successfully using Pandas, scikit-learn and Bokeh. The Iris dataset was loaded and organized into a DataFrame containing petal length, petal width and species. The interactive scatter plot clearly demonstrated a strong positive relationship between petal length and petal width and visually separated the three Iris species. Bokeh interaction tools further enabled hovering, zooming, panning and legend-based filtering, making the visualization suitable for exploratory analysis.

References

1. Bokeh Documentation: [Open Link](#)
2. Pandas Documentation: [Open Link](#)
3. scikit-learn Iris Dataset Documentation: [Open Link](#)
4. Python Program (GitHub link):
5. Class lecture link: [Open Lecture Link](#)

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Program S. No. 14

Title	Interactive Line Plot with Bokeh			
Problem Statement	Create an interactive line plot showing the daily temperatures over a week using Bokeh.			
Objectives	• To understand interactive line visualization using Bokeh. • To organize daily temperature values in a Pandas DataFrame. • To visualize day-wise temperature changes over one week. • To identify the highest, lowest and average weekly temperature. • To use hover, zoom, pan, reset and save tools for interactive time-series analysis.			
Dataset URL Link / File	Dataset: Manually defined weekly temperature readings for seven consecutive days.			
	Day		Temperature (°C)	
	Monday		29	
	Tuesday		31	
	Wednesday		30	
	Thursday		32	
	Friday		34	
	Saturday		33	
	Sunday		31	
Functions Details	Sl. No.	Function	Package	Description of function
	1	DataFrame()	Pandas	Creates a tabular dataset from the day and temperature values.
	2	ColumnData Source()	Bokeh	Converts the DataFrame into a Bokeh data source.
	3	figure()	Bokeh	Creates the interactive Bokeh plotting canvas.
	4	line()	Bokeh	Draws the line connecting daily temperatures.
	5	scatter()	Bokeh	Adds visible markers for each day.
	6	HoverTool()	Bokeh	Displays day and temperature values on hover.
	7	add_tools()	Bokeh	Adds the hover tool to the figure.
	8	idxmax()	Pandas	Locates the day with the maximum temperature.
	9	idxmin()	Pandas	Locates the day with the minimum temperature.
	10	mean()	Pandas	Calculates the average weekly temperature.
	11	output_file()	Bokeh	Defines the HTML output filename.
	12	save()	Bokeh	Saves the interactive line plot as an HTML file.

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Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import Pandas and the required Bokeh modules. 2. Define the seven days of the week and their corresponding temperature values. 3. Create a Pandas DataFrame and display it for verification. 4. Calculate the highest, lowest and average temperature values. 5. Convert the DataFrame into a Bokeh ColumnDataSource. 6. Create a Bokeh figure using the days as categorical X-axis values. 7. Plot the temperature trend using a line and add markers for each day. 8. Add HoverTool to display the exact day and temperature interactively. 9. Save the completed visualization as an HTML file and verify successful execution. Flow: START -> Create Weekly Dataset -> DataFrame -> Temperature Statistics -> ColumnDataSource -> Bokeh Figure -> Line + Markers -> Hover -> Save HTML -> STOP
Program	<pre>import pandas as pd from bokeh.plotting import figure, output_file, save from bokeh.models import ColumnDataSource, HoverTool # Weekly temperature dataset days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"] temperatures = [29, 31, 30, 32, 34, 33, 31] df = pd.DataFrame({"Day": days, "Temperature (°C)": temperatures}) print(df) highest_row = df.loc[df["Temperature (°C)"].idxmax()] lowest_row = df.loc[df["Temperature (°C)"].idxmin()] average_temp = df["Temperature (°C)"].mean() print("Highest Temperature:", highest_row["Day"], "-", highest_row["Temperature (°C)"], "°C") print("Lowest Temperature:", lowest_row["Day"], "-", lowest_row["Temperature (°C)"], "°C") print("Average Temperature:", round(average_temp, 2), "°C") source = ColumnDataSource(df) p = figure(x_range=days, title="Interactive Line Plot: Daily Temperature Over a Week", x_axis_label="Day", y_axis_label="Temperature (°C)", width=900, height=520, tools="pan,wheel_zoom,box_zoom,reset,save")</pre>

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```
p.line(x="Day", y="Temperature (°C)", source=source, line_width=3)
p.scatter(x="Day", y="Temperature (°C)", source=source, size=10)
```

```
hover = HoverTool(tooltips=[
    ("Day", "@Day"),
    ("Temperature", "@{Temperature (°C)} °C")
])
p.add_tools(hover)
p.xgrid.grid_line_color = None
```

```
output_file("program14_temperature_line.html")
save(p)
print("Interactive line plot saved successfully.")
```

**Output
Snapshots
With
Explanation/
Observations**

The following outputs were generated from the completed Program 14. The chart image is a static view of the interactive Bokeh line plot, while the saved HTML output contains the full interactive tools.

Jupyter Output - Weekly Temperature Summary

Weekly Temperature Dataset:
Monday 29 °C | Tuesday 31 °C | Wednesday 30 °C
Thursday 32 °C | Friday 34 °C | Saturday 33 °C | Sunday 31 °C
Highest Temperature: Friday - 34 °C
Lowest Temperature: Monday - 29 °C
Average Temperature: 31.43 °C

Fig. 2.1: Weekly Temperature Dataset and Statistical Summary

The dataset contains seven daily readings. Friday records the highest temperature of 34 °C, Monday records the lowest temperature of 29 °C, and the weekly average is approximately 31.43 °C.

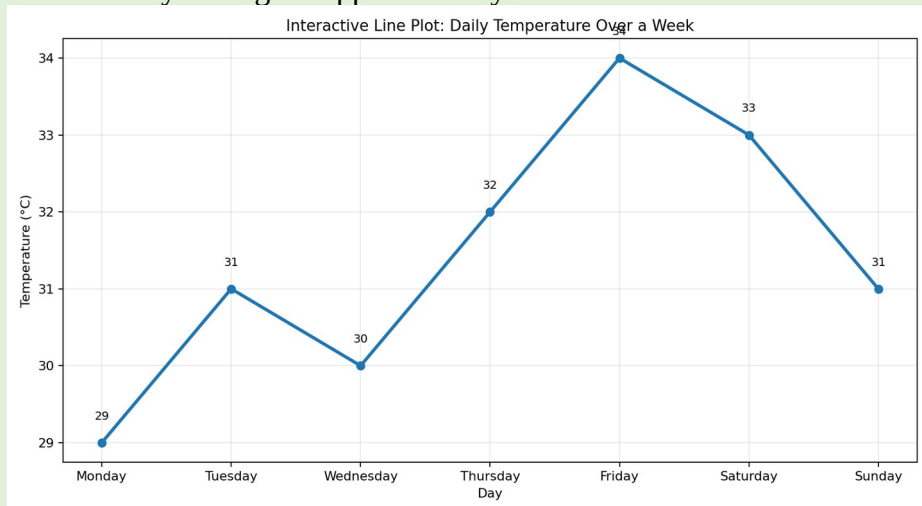


Fig. 2.2: Interactive Line Plot Output

The line plot shows the day-wise temperature variation across the week. The temperature rises overall from Monday to Friday, reaches its maximum on Friday, and decreases during the weekend.

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	Jupyter Output - Program 14 Final Status Interactive line plot created successfully using Bokeh. HTML file saved: program14_temperature_line.html Days visualized: 7 Hover, zoom, pan, reset and save tools are enabled. Fig. 2.3: Program 14 Execution and HTML Save Confirmation The execution confirms successful creation and saving of the Bokeh line plot. The resulting HTML file supports hover, zoom, pan, reset and save interactions.
Conclusions	The exercise was completed successfully using Pandas and Bokeh. Daily temperature values for one week were organized in a DataFrame and visualized using an interactive line plot. Statistical analysis identified Friday as the hottest day, Monday as the coolest day, and the average weekly temperature as approximately 31.43 °C. The interactive Bokeh output makes the weekly trend easy to inspect using hover and navigation tools.
References	1. Bokeh Documentation: Open Link 2. Pandas Documentation: Open Link 3. Python Program (GitHub link): 4. Class lecture link: Open Lecture Link

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Program S. No. 15

Title	Interactive Bar Chart with Bokeh			
Problem Statement	Create an interactive bar chart showing the sales figures of different products using Bokeh.			
Objectives	• To understand interactive categorical bar-chart visualization using Bokeh. • To organize product-wise sales values in a Pandas DataFrame. • To compare sales performance across different products. • To calculate total, highest and average product sales. • To use Bokeh hover, zoom, pan, reset and save tools for interactive analysis.			
Dataset URL Link / File	Dataset: Manually defined product-wise sales values for six products.			
	Product		Sales (Units)	
	Laptop		125	
	Smartphone		210	
	Tablet		95	
	Headphones		170	
	Smartwatch		140	
	Camera		80	
Functions Details	Sl. No.	Function	Package	Description of function
	1	DataFrame()	Pandas	Creates a structured product-sales DataFrame.
	2	ColumnData Source()	Bokeh	Converts the DataFrame into a Bokeh data source.
	3	figure()	Bokeh	Creates the categorical plotting canvas.
	4	vbar()	Bokeh	Draws one vertical bar for each product.
	5	HoverTool()	Bokeh	Displays product and sales values on hover.
	6	add_tools()	Bokeh	Adds the hover tool to the figure.
	7	idxmax()	Pandas	Locates the product with the highest sales.
	8	max()	Pandas	Returns the highest sales value.
	9	sum()	Pandas	Calculates the total number of units sold.
	10	mean()	Pandas	Calculates the average sales per product.
	11	output_file()	Bokeh	Defines the interactive HTML output filename.
	12	save()	Bokeh	Saves the completed Bokeh bar chart as HTML.

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Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import Pandas and the required Bokeh modules. 2. Define the product names and their corresponding sales figures. 3. Create and display a Pandas DataFrame for verification. 4. Calculate total sales, the highest-selling product and average sales. 5. Convert the DataFrame to a Bokeh ColumnDataSource. 6. Create a categorical Bokeh figure with product names on the X-axis. 7. Use vbar() to display the product-wise sales values. 8. Add HoverTool so exact sales values are visible interactively. 9. Save the completed interactive bar chart as an HTML file and verify successful execution. Flow: START -> Create Product Dataset -> DataFrame -> Sales Statistics -> ColumnDataSource -> Bokeh Figure -> Vertical Bars -> Hover -> Save HTML -> STOP
Program	<pre>import pandas as pd from bokeh.plotting import figure, output_file, save from bokeh.models import ColumnDataSource, HoverTool # Product sales dataset products = ["Laptop", "Smartphone", "Tablet", "Headphones", "Smartwatch", "Camera"] sales = [125, 210, 95, 170, 140, 80] df = pd.DataFrame({"Product": products, "Sales (Units)": sales}) print(df) highest_row = df.loc[df["Sales (Units)"].idxmax()] total_sales = df["Sales (Units)"].sum() average_sales = df["Sales (Units)"].mean() print("Total Sales:", total_sales, "units") print("Highest Sales:", highest_row["Product"], "-", highest_row["Sales (Units)"], "units") print("Average Sales:", round(average_sales, 2), "units") source = ColumnDataSource(df) p = figure(x_range=products, title="Interactive Bar Chart: Product Sales", x_axis_label="Product", y_axis_label="Sales (Units)", width=900, height=520, tools="pan,wheel_zoom,box_zoom,reset,save") p.vbar(x="Product", top="Sales (Units)", source=source, width=0.65)</pre>

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```
hover = HoverTool(tooltips=[
    ("Product", "@Product"),
    ("Sales", "@{Sales (Units)} units")
])
p.add_tools(hover)
p.xgrid.grid_line_color = None

output_file("program15_product_sales_bar.html")
save(p)
print("Interactive bar chart saved successfully.")
```

**Output
Snapshots
With
Explanation/
Observations**

The following outputs were generated by executing Program 15. The static chart image presents the product-wise sales comparison, while the generated Bokeh HTML file preserves the interactive functionality.

Jupyter Output - Product Sales Summary

```
Product Sales Dataset:
Laptop 125 | Smartphone 210 | Tablet 95
Headphones 170 | Smartwatch 140 | Camera 80
Total Sales: 820 units
Highest Sales: Smartphone - 210 units
Average Sales: 136.67 units
```

Fig. 3.1: Product Sales Dataset and Statistical Summary

The product dataset contains six sales values. Total sales are 820 units, Smartphone has the highest sales at 210 units, and average sales are approximately 136.67 units per product.

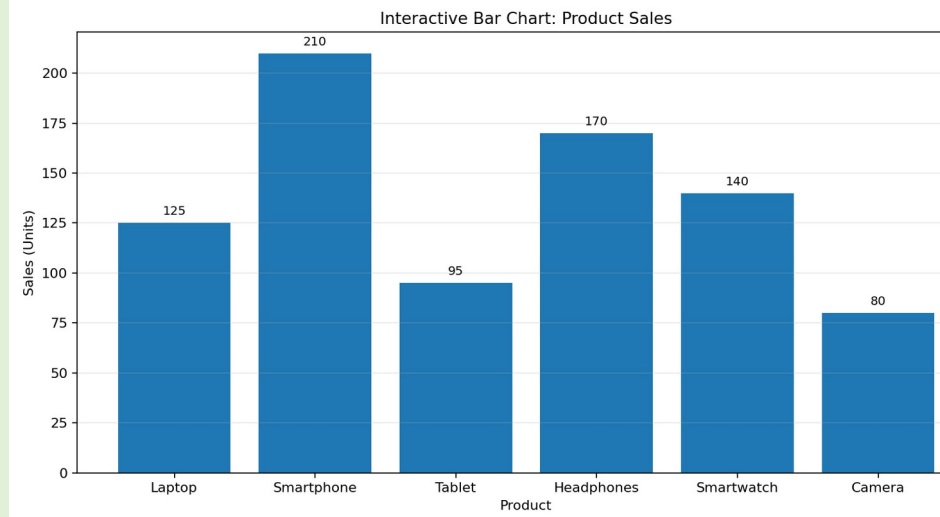


Fig. 3.2: Interactive Bar Chart Output

The bar chart compares sales across all six products. Smartphone is the highest-selling product, followed by Headphones, while Camera records the lowest sales.

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	Jupyter Output - Program 15 Final Status Interactive bar chart created successfully using Bokeh. HTML file saved: program15_product_sales_bar.html Products visualized: 6 Hover, zoom, pan, reset and save tools are enabled. Fig. 3.3: Program 15 Execution and HTML Save Confirmation The final execution confirms successful generation and saving of the Bokeh product-sales bar chart. The HTML output supports hover and navigation tools for interactive inspection.
Conclusions	The exercise was completed successfully using Pandas and Bokeh. Product sales data were organized in a DataFrame and visualized through an interactive vertical bar chart. Smartphone recorded the highest sales with 210 units, total sales were 820 units, and average sales were approximately 136.67 units. The visualization provides a clear product-wise comparison and the interactive Bokeh tools improve data exploration.
References	1. Bokeh Documentation: Open Link 2. Pandas Documentation: Open Link 3. Python Program (GitHub link): 4. Class lecture link: Open Lecture Link

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